Virtual Patient Digital Twin

Predicting health outcomes from EHR and wearable data

Team CardioSync: Sarthak Sonkar, Supriya Suman, Sonam Kumari

Happiest Health Digital Twin Challenge | Build. Predict. Reimagine Healthcare.

Care happens between visits

1

Snapshots, not stories

A clinic visit captures a few numbers. Risk builds quietly in the weeks in between.

2

Data lives in silos

EHR records and wearable streams sit in separate systems, so no one sees the full picture.

3

Advice is generic

Without a model of the individual, guidance cannot be tested before it is prescribed.

The challenge: build a virtual patient model that predicts health outcomes using EHR and wearable data.

A virtual patient you can test plans on

P

Predict

Estimates each patient's 90-day risk of a cardiometabolic event and bands it Low, Moderate or High.

E

Explain

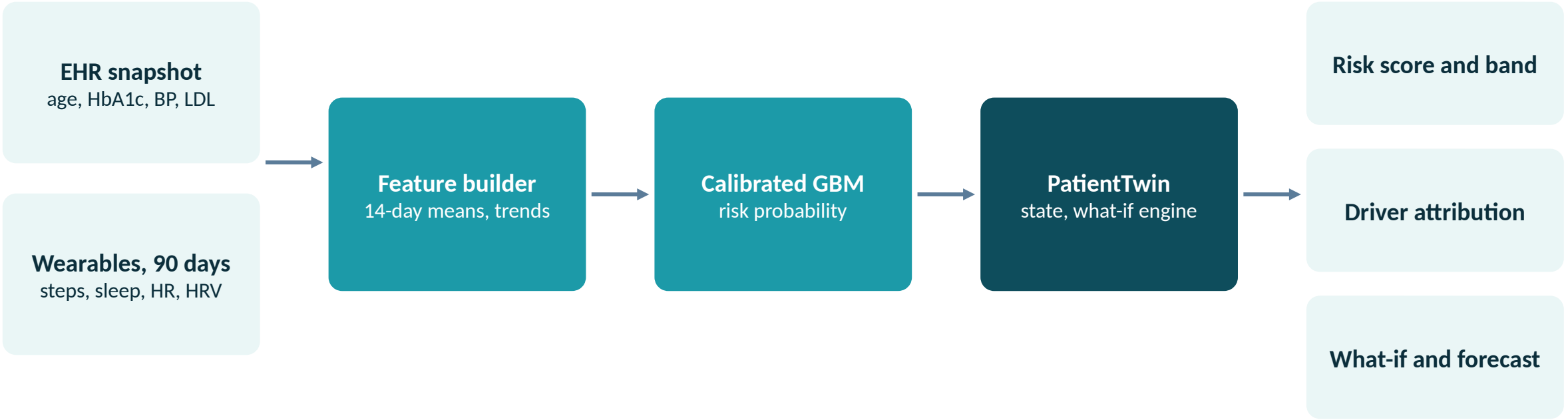
Ranks the modifiable drivers by how much risk would fall if each moved to a healthy reference value.

S

Simulate

Tests what-if plans (steps, sleep, adherence, quitting smoking, weight loss) and forecasts 12 weeks of risk.

How the twin works



Streamlit dashboard: pick a patient, adjust the plan, watch risk change

Data: EHR meets wearables

EHR snapshot

- Age, sex, BMI
- HbA1c, systolic BP, LDL
- Diabetes and smoking status
- On medication and adherence

Wearable stream (90 days)

- Steps, sleep hours, SpO2
- Resting heart rate and HRV
- 14-day means and week-on-week trends
- Sleep variability

Synthetic by design: 1,500 patients, 135,000 daily wearable records, no real patient data.

Results on held-out data

0.86

AUROC on a held-out 25% test set

0.058

Brier score (lower is better)

1,500

synthetic patients modelled

8.1%

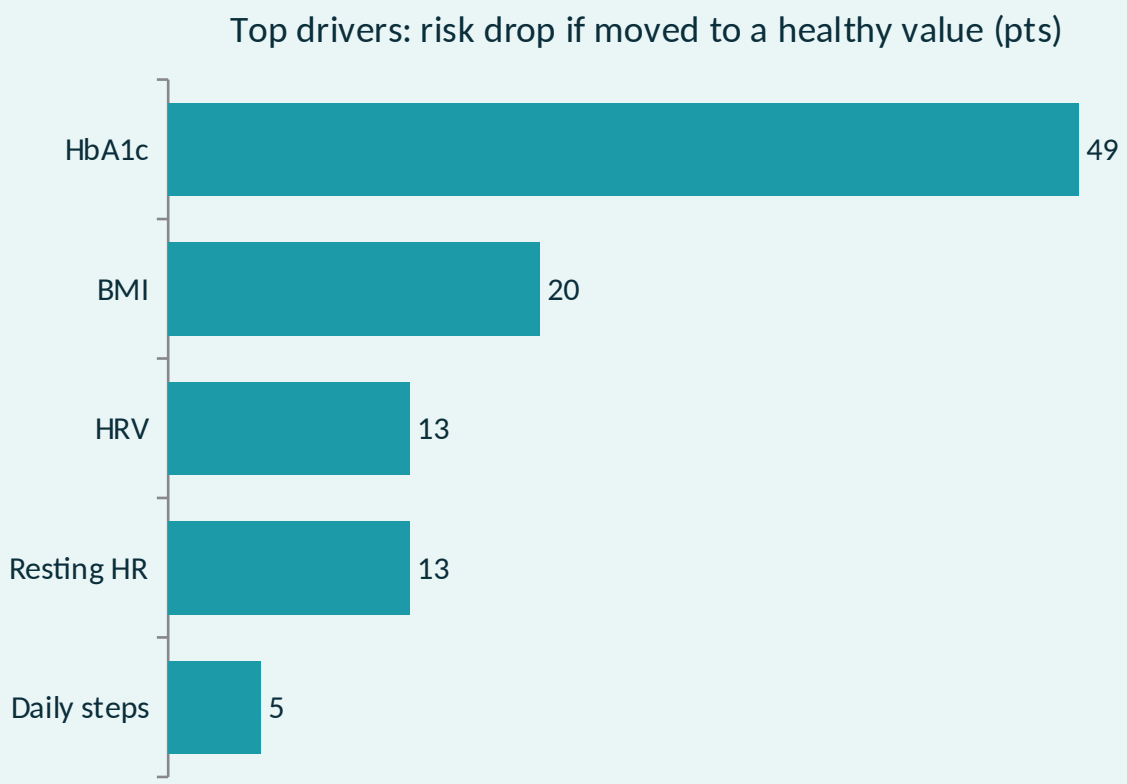
90-day event rate in the data

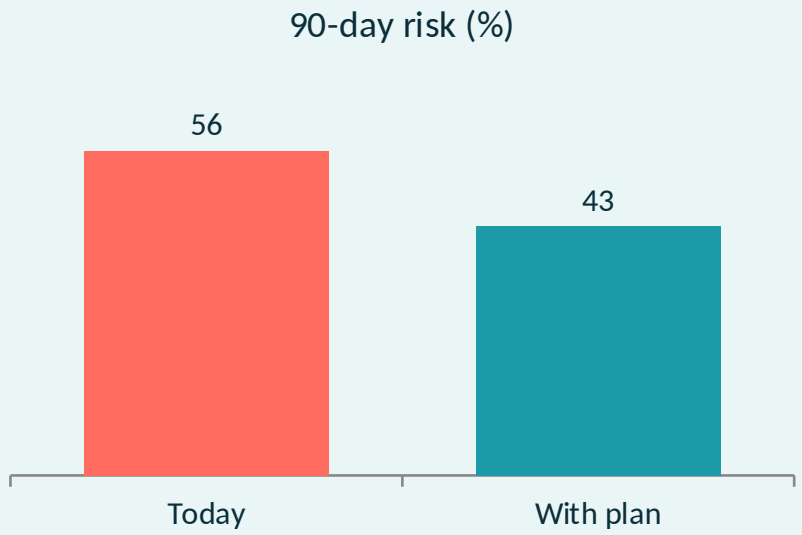
Model

Gradient boosting with isotonic probability calibration, trained on a 75/25 stratified split. Calibration matters because the twin reports risk as a probability.

Honest caveat: synthetic data validates the pipeline, not clinical accuracy.

Example twin: 56% risk drops to 43%





The plan

+3,000 steps a day, +1 hour of sleep, quit smoking

Drivers are one-at-a-time counterfactuals, so they do not add up to the total.

Who benefits

1

Clinicians

Spot rising risk between visits and focus time on the patients who need it most.

2

Health coaches

Show which habit change moves a patient's risk the most, before the first session.

3

Patients

See their own risk fall as steps, sleep and adherence improve. Motivation with evidence.

4

Health systems

Shift from reactive treatment to prevention and reduce avoidable admissions.

Limits, ethics and what's next

Limits and ethics

- Trained on synthetic data, so accuracy will not transfer to real patients
- What-if effects are documented assumptions, not causal estimates
- Decision support only; a clinician reviews every recommendation
- Real use needs consent, de-identification and a bias audit

Roadmap

- FHIR ingestion for real EHR data
- Apple Health and Fitbit wearable connectors
- Temporal twin (state-space or LSTM) for individual trajectories
- SHAP explanations and a fairness audit
- Clinician-in-the-loop review workflow

Thank you

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Code, README and docs are in the GitHub repository.